MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 8

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1. Determine the biggest term of the sequence given by

   (a) $f_n = \dfrac{\sqrt{n}}{n + 2241}$

   (b) $g_n = \frac{n^2}{n^3+200}$

2. Prove the uniform continuity of the following functions by definition.

   (a) $f(x) = x^2 - 3x - 1$, $x \in [3, 6]$

   (b) $f(x) = 2x - 1$, $x \in \mathbb{R}$

3. Prove that the function $f(x) = \frac{|\sin x|}{x}$, $x \neq 0$, is uniformly continuous on each of the intervals $(-1, 0)$ and $(0, 1)$, but not on their union $(-1, 0) \cup (0, 1)$.

4. Use the Mean Value Theorem to show that $|\sin w - \sin z| \leq |w - z|$ for $0 \leq z < w \leq 2\pi$.

5. Suppose $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Suppose $f(a) = 0$ and $|f'(x)| \leq A|f(x)|$ for a constant A.

   (a) Prove that $f(x) = 0$ on $\left[a, a + \dfrac{1}{2A}\right]$.

   (b) Prove that $f(x) = 0$ on the whole interval $[a, b]$.

— End —